Accelerating Adaptive Parallel Tempering with FPGA-based p-bits

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***Abstract*—Special-purpose hardware to solve optimization problems formulated as Ising models has generated great excitement recently. Despite a large diversity in hardware, most solvers employ standard variations of the classical (simulated) annealing (CA) algorithm. Here, we show how powerful replica-based Parallel Tempering (PT) algorithms can significantly outperform CA, using FPGA-based probabilistic computers. Using a massively parallel (graph-colored) architecture, we implement the Adaptive PT (APT) algorithm, generating problem-dependent temperature profiles to equalize replica swap probabilities. We benchmark our p-computer against analytical results from classical Ising theory and use our machine to solve spin-glass instances formulated as hard optimization problems. APT outperforms heuristic choices of temperature profiles used in conventional PT and a replica-based version of CA. Our machine provides 6,000X speedup over optimized CPU, with orders of magnitude further speedup projected for scaled implementations. The developed co-design techniques may be useful for a broad range of Ising machines beyond p-computers.**

Building dedicated hardware to solve Ising models in the beyond-Moore era of computing has generated great excitement [1]. One of the main applications of such quantum-inspired solvers has been hard combinatorial optimization. Despite a diversity of hardware approaches, most solvers use standard algorithms such as classical (simulated) annealing (CA) or direct Markov Chain Monte Carlo (MCMC) sampling without any annealing [1, 2]. In this paper, we show how a family of powerful replica-based algorithms, also known as Parallel Tempering (PT) [3], significantly outperform single-replica CA using an FPGA implementation of a probabilistic computer (FIG. 1a). The key idea of PT is to construct r identical replicas of a given graph and operate each replica at a different (algorithmic) temperature. Occasionally the replicas compare their states and if a higher temperature replica carries a lower energy state, this state is swapped with the lower temperature. The intuition behind the PT algorithm is to avoid terminally getting stuck in a suboptimal state as in the case of single-replica CA where temperature is gradually reduced. Instead, PT ensures that all replicas operate at a constant temperature and the possibility of swaps serves as a means of escaping local minima efficiently (FIG. 1b). One choice of the swap probability between two replicas operating at inverse temperatures β_i and β_j with energies E_i and E_j is:

$$p_a = \min\{1, \exp[\Delta\beta(E_i - E_j)]\} \tag{1}$$

APT algorithm: Just as the β profile in time for CA may need to be optimized, the β profile in space for PT needs to be carefully adjusted. Heuristic choices of β_i over replicas may cause inefficient swap attempts, creating a bottleneck and affecting solution quality. An efficient approach is to maintain a *constant* p_a across each replica pair. Adaptive parallel tempering (APT) is a powerful PT algorithm that optimizes the β_i profile and number of replicas (r) in a *problem-dependent* manner (FIG. 1c). The APT prescription is as follows [4]:

$$\beta_{i+1} = \beta_i + \alpha/\sigma \tag{2}$$

where α is a hyperparameter and σ^2 is the β dependent energy variance in a replica.

Hardware architecture: We have implemented our p-computer on a Xilinx Virtex UltraScale+ VU29P FPGA (FIG. 1a). The architecture is massively parallel and uses graph coloring to update p-bit blocks exploiting the sparsity of the graphs [2]. The p-computer is synthesized once on the FPGA for a family of problems and reused to fit different instances.

Benchmarking PT on 2D Ising: We first benchmark our p-computer with a standard linear PT on 2D finite-size Ising lattices (FIG. 2) and compare the experimentally computed average energy and specific heat ($\sigma^2\beta^2$) with the analytical results [5]. We see excellent agreement with analytical theory at different L and β, especially at larger system sizes. This experiment establishes that our PT implementation samples from the Boltzmann distribution despite having pseudorandom numbers and a fixed point precision.

Problem description: Hard 512 p-bit chimera droplet instances [6] (FIG. 3) are solved to evaluate the performance of APT and CA. These instances contain a random background spin-glass and many structured clusters with a power-law distribution of cluster sizes. The clusters have boosted edges, making the instances very hard for probabilistic solvers and annealers.

APT vs classical annealing: FIG. 4 shows the main results of the APT algorithm accelerated with p-bits. APT hyperparameters are optimized for only one instance (FIG. 5) and the same parameters are used for all the other instances. A sample run in FIG. 4a,b shows p_a is fairly constant over all replica pairs as intended by APT. Next, we benchmark APT by comparing it to linear PT and replicated CA (RCA) (FIG. 4c). All these replica-based algorithms use the same hardware on the FPGA with the same amount of computation per trial. Both APT and RCA outperform linear PT. We also show single replica CA in the comparison which often times out, proving the strength of replica-based algorithms. RCA requires less wall-clock time and pre-processing, however, APT shows a higher success probability (FIG. 4d). Replica-based algorithms clearly outperform CA and should be preferred if hardware is available to run replicas in parallel.

Projections and outlook with sMTJs: FPGA-based p-computer provides 6,000X acceleration over the optimized graph-colored implementation on CPU (Table I) making it suitable to explore the strength of replica-based algorithms as shown in this work. It fits the best energy-throughput corner compared to other 2D Ising accelerators (FIG. 6). Recent progress and experimental demonstration with fast sMTJs project a further 1,000X improvement for high-performance and energy-efficient probabilistic computing in scaled implementations, illustrating the potential of the approach.

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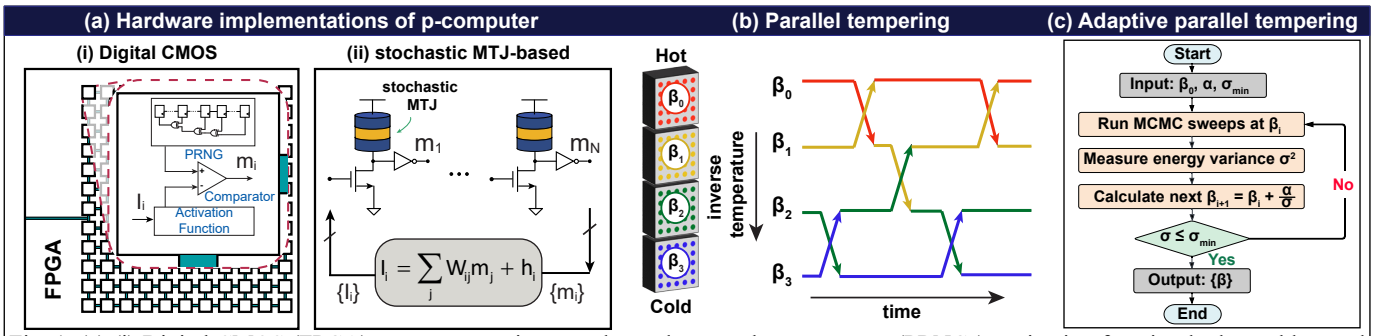

Fig. 1. (a) (i) Digital CMOS (FPGA) p-computer using pseudo-random number generators (PRNGs), activation function lookup tables, and comparators up to ~10,000 p-bits. (ii) Nanodevice p-computer with sMTJs projected up to ~1M p-bits. (b) Parallel tempering (PT) algorithm: multiple p-bit replicas are run at different inverse temperatures (β) and allowed to swap states randomly. (c) Adaptive parallel tempering (APT) algorithm pre-processes the β schedule adaptively to equate swap probabilities by minimizing energy variance of replicas.

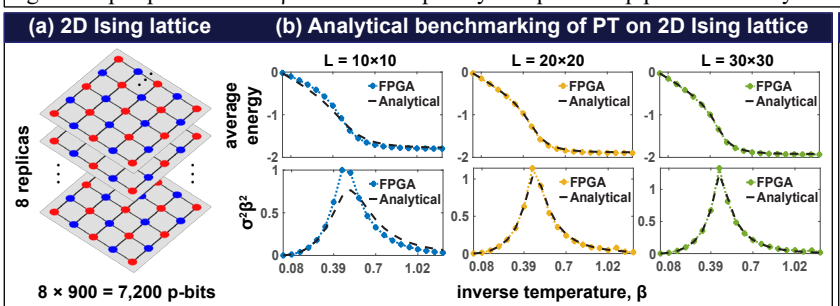

Fig. 2. (a) 2D Ising lattices of different sizes (up to L = 30×30 = 900 p-bits) are used to analytically benchmark the p-computer. 8 replicas with 7,200 p-bits are used to represent 32 values of β. (b) Analytical benchmarking of PT on the 2D Ising lattices. Both the average and the variance (σ^2) of the energy are computed reading 10^6 sweeps at each β where $\beta^2\sigma^2$ is the specific heat capacity. 10^4 PT swaps are attempted among 8 replicas. The results validate PT implementation on p-computer despite using PRNGs and a fixed-point precision (13 bits: 1 sign, 6 integer and 6 fraction bits).

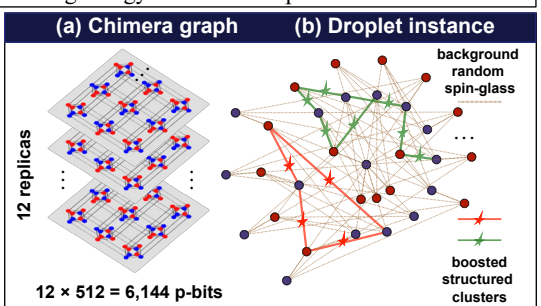

Fig. 3. (a) Chimera graphs used in this work with 12 × 512 = 6,144 p-bits. (b) Structured droplet instance on a chimera graph. Background random spin-glass is weighted from a random probability distribution. Certain structured clusters are generated and given boosted weights. These are specifically hardened optimization problems to test algorithmic solvers.

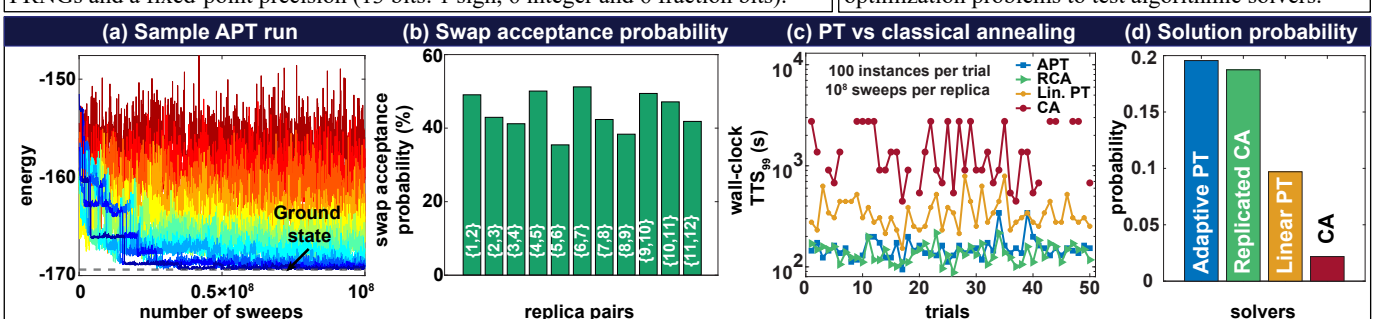

Fig. 4. (a) Sample APT run on the 512 p-bit chimera taking 10^8 sweeps per replica with 1000 swap attempts. Fig. 5 below shows the APT schedule optimization. (b) Swap acceptance probability between each replica pair, a roughly uniform acceptance rate is achieved as desired from the APT. (c) Comparison of linear PT and APT with replicated classical annealing (RCA) and single replica classical annealing (CA). 100 different instances on the 512 p-bit chimera graph are solved per trial taking 10^8 sweeps per replica. Wall-clock time to solution (TTS_{99}) to solve the instances with 99% probability. Missing points in the CA indicate a time-out with no success. For CA/ RCA a linear schedule of β = 0 to 25 with 100 steps is used. For linear PT, we spaced β = 1 to 25 linearly in 12 replicas. (d) Success probability for different solvers.

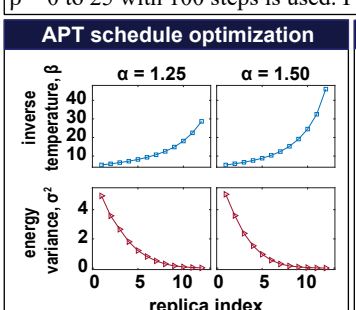

Fig. 5: Starting $\beta_0 \approx 5$ gives a high energy variance. α = 1.25 requires more replicas to minimize σ. But high α = 1.50 reduces the swap acceptance rate by spacing out β. α = 1.25 is used in this work.

Table I: FPGA acceleration

Metric \ Platform	FPGA	CPU (graph-colored)	CPU (standard)
Wall-clock time (s)	7.00	42200	Time-out
Flips/ns	87.77	0.0146	0.000013
Time/sweep (ns)	5.83	35166.67	3.83×10^7

35166 ÷ 5.83 ≈ 6,000X improvement in FPGA over optimized CPU

Table I. Time required to compute an APT trial with 10^8 sweeps per replica and 10^3 swap attempts. Hardware acceleration using FPGA provides 6,000X improvement over the optimized graph-colored CPU and 6,000,000X over the standard CPU.

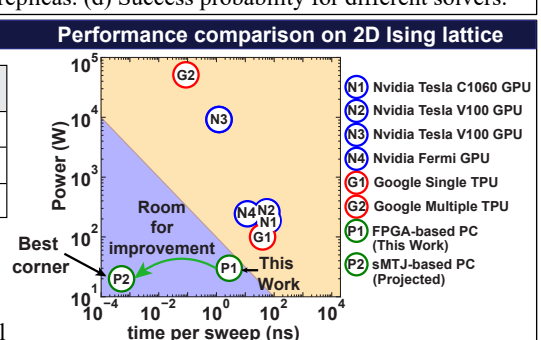

Fig. 6: Comparison of probabilistic accelerators solving 2D Ising lattices. Measured performance of digital FPGA with 7,200 p-bits on 2D Ising is (P1). Projected performance with sMTJs offers further room for improvement (P2).

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